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Information needs of adult childhood, adolescent, and young adult cancer survivors (CAYACS): A systematic review

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ABSTRACT

Objective: Childhood, adolescent, and young adult cancer survivors (CAYACS) often report cancer-related knowledge gaps. Addressing their information needs is associated with better quality of life. We aimed to explore and synthesize evidence on CAYACS' cancer-related information needs and identify associated characteristics.

Methods: Peer-reviewed articles on information needs in adult CAYACS ≥ 5 years post-diagnosis were systematically searched in PubMed, PsycINFO, and Scopus. The quality of included publications was assessed using the Mixed Methods Appraisal Tool, and results were narratively synthesized.

Results: Twenty-one studies (n = 10 quantitative, n = 8 qualitative, n = 3 mixed-methods) with a total of 5624 participants (range: 14–1386 per study) were included. Between 51 % and 77 % of CAYACS had at least one information need. Needs were reported across 11 domains, including cancer-related health information (2 %-86 %), follow-up care and prevention (2 %-91 %), healthcare system interactions (9 %-36 %), living a healthy lifestyle (4 %-60 %), psychosocial well-being and support (12 %-40 %), sexual health (<1 %-32 %), finances (2 %-50 %), relationships (2 %-20 %), education and employment (<1 %-18 %), insurances (28 %-47 %), and peer support (7 %-35 %). The highest prevalences were observed in follow-up discussions on current health (92 %) and late effects (19 %-86 %). Female sex, older age at study, lower educational attainment, poorer mental and physical health, longer time since diagnosis/treatment, central nervous system tumor diagnosis, and lack of written information were associated with more information needs.

Conclusions and practice implications: Adult CAYACS report significant information needs years after treatment, particularly regarding cancer-related health information, follow-up care and prevention, and lifestyle. Addressing these needs with age-appropriate, individualized information may improve their quality of life. Electronic and mobile health tools are promising methods to provide such support.

1. Introduction

Patient information is important at every stage of the cancer trajectory. However, it is not always readily available during the later stages, when patients enter survivorship. This is particularly relevant for long-term cancer survivors, who may face ongoing concerns, including psychosocial and physical late effects of cancer and its treatments [1].

Late effects can impact survivors throughout their lives, even years after their diagnosis. They remain at risk for health conditions such as organ dysfunction and second neoplasms [2], or psychosocial problems,

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which can affect their health-related quality of life (HRQOL) [3,4]. Childhood, adolescent, and young adult on-set cancer survivors (CAYACS) may be particularly vulnerable as their cancer or late effects can disrupt normal development including developing independence and establishing a career or family [5–8]. Lifelong multidisciplinary follow-up care is recommended to address their risks of late effects and to improve long-term outcomes [9–11].

A crucial factor in long-term follow-up care (LTFU) is providing information that meets survivors' needs, because it is associated with higher satisfaction with care, better HRQOL, and lower levels of depression and anxiety among long-term survivors [12–14]. Although CAYACS face a high risk of late effects following cancer treatment, many do not adhere to the recommended LTFU based on their risk [15,16]. The decision to attend and adhere to LTFU is associated with CAYACS' knowledge [11,14]. Therefore, empowering them with appropriate information is essential for successful long-term health management.

Research has shown that significant knowledge gaps exist among CAYACS on a range of topics, including late effects, cancer diagnosis and treatment, cancer recurrence, sexuality, lifestyle, and psychosocial issues [1,11,14,17–24]. A Norwegian study among CAYACS showed that 50 %-60 % did not receive information about the disease, treatment and late effects, and consequently reported information needs [14]. Similar results have been reported in other European countries [25,26]. In addition, CAYACS may find it difficult to obtain support for their psychosocial and physical concerns, particularly when transitioning to adult services [1,27]. A serious consequence of unmet needs is suboptimal health outcomes, such as poor management of mental and physical health issues [22,23].

While systematic reviews have been conducted on the information needs of relatives of childhood cancer patients and survivors [28,29], a comprehensive overview of CAYACS' information needs is lacking. Insight into CAYACS' information needs and their associated factors is essential for shaping targeted information programs and guiding interventions. Therefore, this systematic review aimed to: (1) comprehensively explore and synthesize the evidence on cancer-related information needs of adult CAYACS, and (2) identify characteristics associated with these needs.

2. Methods

2.1. Protocol and registration

This systematic review adheres to the PRISMA guidelines for the reporting of systematic reviews and meta-analyses [30]. The review protocol was registered in PROSPERO (registration number: CRD42024505031).

2.2. Search strategy

The literature search was conducted on November 20, 2024, and included peer-reviewed articles on the cancer-related information needs of adult CAYACS. The PubMed, PsycINFO, and Scopus databases were searched using four blocks of search terms: "information needs", "cancer", "survivors", and "childhood, adolescence, young adulthood". The blocks were adapted for each database (Appendix A). When possible, the "humans only" and "peer-reviewed" filters were selected. To provide a comprehensive overview, no restrictions on geographical region and publication year were applied.

2.3. Inclusion and exclusion criteria

We included original peer-reviewed English research articles that aimed to assess information needs as reported by adult CAYACS. The inclusion criteria for the population were as follows (for ≥ 75 % of the study sample): survivors of any type of cancer diagnosed between 0 and 25 years of age, aged ≥ 18 years at study, and ≥ 5 years since diagnosis at the time of the study. The upper limit of 25 years at diagnosis was chosen to focus on survivors diagnosed during childhood, adolescence, or emerging adulthood, as these survivors may have unique needs related to their developmental stage [8]. Survivors diagnosed beyond age 25 may face different challenges related to work, family responsibilities, or established adult roles, and may therefore require different informational support. Studies with qualitative, quantitative, and mixed methods designs were included to gain a comprehensive understanding of CAYACS' information needs. Quantitative studies describe patterns and outcomes, while qualitative studies provide insight into underlying reasons and preferences.

We excluded studies that: focused on diseases other than cancer or did not clearly distinguish between cancer and other diseases; lacked a clear distinction between information needs and other support needs or cancer-related aspects; did not differentiate between CAYACS' and relatives' information needs; did not rely on CAYACS' self-reported information needs; were editorials, conference abstracts, commentaries, dissertations, or protocols; and were secondary research such as systematic reviews, or meta-analyses.

2.4. Study selection

After database searches, references were imported into EndNote 21, where duplicates were removed. The titles and abstracts from the remaining studies were screened in Rayyan by two independent reviewers (EB or AM, and AI or KO). The full texts of potentially eligible studies were retrieved and independently assessed by two reviewers (EB or AM, and AI or KO). Disagreements were resolved through consultation with a third reviewer (GM or AI), if necessary.

2.5. Data extraction

The following data were extracted by the first author (EB) and checked for accuracy by the other first (AM) or second author (KO): study characteristics (first author, title, year, country, aim, study design, method of data collection and analyses, time points of data collection, sample size), study population (gender, age, diagnosis, treatment, age at diagnosis, time since diagnosis), information needs (definition, identified needs with names and prevalence, need status as unmet or met), associated factors and their estimates, and statistical analyses.

Unmet needs were defined as information needs that are currently present and have not yet been (fully) addressed, including a reported interest, need, or desire for receiving more information on a particular topic. Met needs were defined as instances where no (further) information or discussion is needed, or the required information has already been sufficiently or satisfactorily provided. Notably, since most studies did not distinguish between no needs and met needs, these were combined into one single category.

Data were not extracted when needs other than information needs were reported (e.g., support needs), or when CAYACS' needs were combined with those of relatives.

2.6. Quality assessment

The quality of the included studies was assessed using the Mixed Methods Appraisal Tool (MMAT), version 2018. This tool allows for the evaluation of the methodological quality of systematic reviews that include qualitative, quantitative, and mixed methods studies [31,32]. It includes two screening questions for all studies and a comprehensive set of criteria for each study design, rated as 'yes', 'no', or 'can't tell' [31]. The full criteria can be found in Appendix B. Consequently, an overall quality score was assigned, with 20 % allocated per criterion met, i.e., 0 % indicating no criteria met and 100 % indicating all criteria met [31] (Table 1). The quality assessment was performed by the first authors (EM or AM) and checked for accuracy by the second author (KO). The aim of the assessment was to evaluate the quality of each study, ensuring

Table 1
Study characteristics.

First author Year Country	Title	Methodology Design Data collection method Needs assessment tool[1]	Area of assessed information needs[2]	Participants Gender	Cancer diagnosis	Age at diagnosis (years)[3]	Time since diagnosis (years)[3]	Quality Appraisal Score (MMAT)[6]
Baudry 2024 France [51]	Supportive care needs of adolescents and young adults 5 years after cancer: a qualitative study	Qualitative Cross-sectional Interviews	Concern for cancer repercussions, follow-up, cancer relapse, linked to oncological healthcare pathway, about existing support or organizations	Survivors: n = 15, Male: n = 4, Female: n = 8	Lymphoma = 6 (40 %), CNS Tumor = 2 (13.3 %), Bone tumor = 4 (26.7 %), Soft-tissue sarcoma = 2 (13.3), Germ cell and gonadal tumor = 1 (6.7)	Mean = 15 (SD 1.73), Median = 14 (Range 12–18)	5 years post diagnosis	100 %
Billman 2023 USA [35]	Perspectives of childhood cancer survivors as young adults: a qualitative study of illness education resources and unmet information needs	Qualitative Cross-sectional Interviews	Illness education resources	Survivors: n = 14 Male: n = 6 Female: n = 8	Leukemia/lymphoma: n = 8 Solid tumor: n = 6	≤ 10	-	100 %
Christen 2018 Switzerland [36]	Perceived information provision and information needs in adolescent and young adult cancer survivors	Quantitative Cross-sectional Survey Self-developed	Disease, treatment, follow-up, late effects	Survivors: n = 160 Male: n = 98 Female: n = 62	CNS tumor: n = 15 Germ cell tumor: n = 46 Other tumor: n = 26	M = 21.6 (SD 2.9)	M = 12.4 (SD 4.8)	60 %
Claridy 2018 USA, Canada [37]	Patterns of internet-based health information seeking in adult survivors of childhood cancer	Quantitative Cross-sectional Survey Self-developed	Health and medical information	Survivors: n = 1386 Male: n = 699 Female: n = 687	Leukemia: n = 211 Hodgkin lymphoma: n = 129 Non-Hodgkin lymphoma: n = 181 CNS tumor: n = 171 Neuroblastoma: n = 159 Renal tumor: n = 188 Bone tumor: n = 148 Soft tissue sarcoma: n = 198	0–4: n = 595 5–9: n = 305 10–14: n = 275 15–20: n = 211	15–19: n = 275 20–24: n = 548 25–29: n = 320 30 + : n = 243	80 %
Cox 2016 USA [38]	The unmet emotional, care/support, and informational needs of adult survivors of pediatric malignancies	Quantitative Cross-sectional Survey (CCSS-NAQ)	9 domains (psycho-emotional, healthcare system concerns, cancer-related health information, general health, survivor care and support, surveillance-related information, coping, fiscal concerns, and relationships)	Survivors: n = 1189 Male: n = 465 Female: n = 724	Leukemia: n = 401 Hodgkin lymphoma: n = 151 Non-Hodgkin lymphoma: n = 88 CNS tumor: n = 159 Neuroblastoma: n = 76 Wilms' tumor: n = 121 Bone cancer: n = 86 Soft-tissue sarcoma: n = 107	0–4: n = 452 5–9: n = 271 10–14: n = 247 15–20: n = 219	24–27: n = 311 28–31: n = 295 32–35: n = 293 36–42: n = 290	80 %
D'Agostino 2013 Canada [39]	Psychosocial challenges and resource needs of young adult cancer survivors: implications for program development	Qualitative Cross-sectional Focus group	Not specified	Survivors: n = 22 Male: n = 12 Female: n = 10	Leukemia: n = 5 Brain tumor: n = 9 Breast tumor: n = 3 Other tumor: n = 5	4–29	M = 11.9 Range = 2–27[4]	100 %
Frederick 2017 USA [52]	Preparing childhood cancer survivors for transition to adult care: the young adult perspective	Qualitative Cross-sectional Focus group	Cancer history, late effects, follow-up, barriers and facilitators to transition	Survivors: n = 16 Male: n = 4 Female: n = 12	Leukemia: n = 4 Hodgkin lymphoma: n = 1 Non-Hodgkin lymphoma: n = 1 Brain tumor: n = 5 Retinoblastoma: n = 1 Renal tumor: n = 1 Bone tumor: n = 2 Other: n = 1	M = 8.5 (SD 4.9) Range = 0–17	Time since completion of therapy: < 5 years: n = 2 5–10 years: n = 3 > 10 years: n = 11	100 %

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Table 1 (*continued*)

First author Year Country	Title	Methodology Design Data collection method Needs assessment tool[1]	Area of assessed information needs[2]	Participants Gender	Cancer diagnosis	Age at diagnosis (years)[3]	Time since diagnosis (years)[3]	Quality Appraisal Score (MMAT)[6]
Gianinazzi 2014 Switzerland [40]	Information provision and information needs in adult survivors of childhood cancer	Quantitative Longitudinal Survey Self-developed	Disease, treatment, follow-up, late effects	Survivors: n = 319 Male: n = 140 Female: n = 179	Leukemia: n = 115 (36 %) Lymphoma: n = 59 (18 %) CNS tumor: n = 38 (11 %) Neuroblastoma: n = 8 (4 %) Retinoblastoma: n = 5 (2 %) Renal tumor: n = 21 (6 %) Hepatic tumor: n = 1 (1 %) Bone tumor: n = 22 (7 %) Soft tissue sarcoma: n = 20 (6 %) Germ cell tumor: n = 9 (3 %) Other tumor: n = 8 (2 %) Langerhans cell Histiocytosis: n = 13 (4 %)	M = 8.9 (SD 4.7)	M = 12.5 (SD 3.8)	60 %
Hendriks 2020 Switzerland [53]	The unmet needs of childhood cancer survivors in long-term follow-up care: a qualitative study	Qualitative Cross-sectional Interviews	Not specified	Survivors: n = 28 Male: n = 9 Female: n = 19	Leukemia: n = 10 Lymphoma: n = 5 CNS tumor: n = 3 Renal tumor: n = 3 Bone tumor: n = 5 Soft tissue sarcoma: n = 2	0–5: n = 7 6–11: n = 10 12–18: n = 11 M = 9.3	Time since end of treatment: ≤ 5: n = 3 6–15: n = 9 16–25: n = 6 > 25: n = 10 M = 19.1	100 %
Hovén 2018 Sweden [41]	Information needs of survivors and families after childhood CNS tumor treatment: a population-based study	Quantitative Cross-sectional Survey (Self-developed based on the EORTC QLQ-INFO26 inventory)	Illness and treatment, medical tests, monitoring of late effects, psychological support, and rehabilitation services	Survivors: n = 518 Male: n = 263 Female: n = 255	Astrocytoma/glial tumor: n = 276 Ependymoma: n = 43 PNET/ Medulloblastoma: n = 66 Other CNS tumor: n = 133	< 5: n = 70 5–10: n = 146 10–15: n = 213 > 15: n = 89 M = 10.6 Mdn = 11.3 (SD 4.4)	< 10: n = 89 10–15: n = 129 15–20: n = 178 > 20: n = 122 M = 15.6 Mdn = 16.1 (SD 5.0)	100 %
Jardim 2021 Brazil [42]	Fertility-related concerns and uncertainties in adolescent and young adult childhood cancer survivors	Qualitative Cross-sectional Interviews	Fertility-related concerns	Survivors: n = 24 Male: n = 13 Female: n = 11	Leukemia: n = 8 Lymphoma: n = 3 Brain tumor: n = 5 Solid tumor: n = 7 Adrenal tumor: n = 1	M = 8.1 (SD 3.8) Range = 1–15	Cancer survival time: M = 10.4 (SD 3.2) Range = 5–15	100 %
Kelada 2019 Australia, New Zealand [43]	Perceived cancer-related pain and fatigue, information needs, and fear of cancer recurrence among adult survivors of childhood cancer	Quantitative Cross-sectional Survey Self-developed	Pain and fatigue management	Survivors: n = 404 Male: n = 174 Female: n = 228	Leukemia: n = 172 Lymphoma: n = 58 CNS tumor: n = 58 Neuroblastoma: n = 21 Wilms' tumor: n = 24 Bone and soft tissue sarcoma: n = 50 Other: n = 19	< 16	M = 19.1 (SD 8.1) Range = 5–59	60 %
Lie 2015 Norway [44]	Providing information about late effects after childhood cancer: Lymphoma survivors' preferences for what, how and when	Qualitative Cross-sectional Focus group	Late effects	Survivors: n = 34 Male: n = 15 Female: n = 19	Hodgkin lymphoma: n = 21 Non-Hodgkin lymphoma: n = 13	M = 13 (SD 4) Range = 3–18	M = 26 (SD 9) Range = 12–39	100 %
Mayes 2016 United	Health promotion and information provision during long-term follow-up	Qualitative Cross-sectional Interviews	Future health risks, disease prevention, health promotion	Survivors: n = 51 Male: n = 25	Leukemia: n = 12 Hodgkin lymphoma: n = 10 CNS tumor: n = 6	Mdn = 11.3 Range = 2–22	Time since end of treatment: Mdn = 10 Range = 5–19	40 %

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Table 1 (*continued*)

First author Year Country	Title	Methodology Design Data collection method Needs assessment tool[1]	Area of assessed information needs[2]	Participants Gender	Cancer diagnosis	Age at diagnosis (years)[3]	Time since diagnosis (years)[3]	Quality Appraisal Score (MMAT)[6]
Kingdom [45]	for childhood cancer survivors: a service evaluation			Female: n = 26	Extracranial tumor: n = 23			
McClellan 2013 USA [46]	Understanding the functional late effects and informational needs of adult survivors of childhood cancer	Mixed methods Cross-sectional Survey with space to write additional information about experiences Self-developed	Not specified	Survivors: n = 271/272[5] Male: n = 126 Female: n = 144 Missing: n = 2	Leukemia/lymphoma: n = 130 Brain tumor: n = 51 Solid tumor: n = 90 Missing: n = 1	M = 10.2 (SD 5.2)	-	80 %
Michel 2009 United Kingdom [54]	Follow-up care after childhood cancer: survivors' expectations and preferences for care	Quantitative Longitudinal Survey Self-developed	Medical topics, psychosocial needs, lifestyle behaviors	Survivors: n = 112 Male: n = 49 Female: n = 63	Leukemia: n = 44 Lymphoma: n = 21 CNS tumor: n = 14 Other solid tumor: n = 33	M = 7.0 (SD 4.5) Range = 0.1–16	M = 21.3 (SD 7.9) Range = 5.9–36.1	100 %
Nicklin 2022 United Kingdom [47]	Long-term unmet supportive care needs of teenage and young adult (TYA) childhood brain tumor survivors and their caregivers: a cross-sectional survey	Quantitative Cross-sectional Validated questionnaire (SCNS-SF34)	Not specified	Survivors: n = 69 Male: n = 37 Female: n = 32	Medulloblastoma: n = 24 Astrocytoma: n = 18 Craniopharyngioma: n = 6 Pineal tumor: n = 4 Choroid plexus carcinoma: n = 4 Ependymoma: n = 3 Other: n = 10	0–4: n = 22 5–10: n = 31 11–14: n = 16 M = 7.2	M = 17.4 (SD 4.9) Range = 7–27	40 %
Park 2018 Canada [48]	Unmet needs of adult survivors of childhood cancers: associations with developmental stage at diagnosis, cognitive impairment, and time from diagnosis	Quantitative Cross-sectional Survey (SUNS with self-developed Psychosocial and Practical Unmet Needs Scales)	Not specified	Survivors: n = 115 Male: n = 47 Female: n = 68	Lymphoma and reticuloendothelial system: n = 34 Leukemia: n = 15 CNS: n = 12 Neuroblastoma: n = 3 Renal tumor: n = 3 Wilms' tumor: n = 4 Bone/Connective tissue: n = 14 Testicular: n = 6 Gynecologic: n = 3 Epithelial neoplasm: n = 6 Thyroid: n = 9 Head and neck: n = 3 Other: n = 3	M = 11.5 (SD 5.7)	M = 23.3 (SD 12.2) Mdn = 21 Range = 3.7 – 62.9	60 %
van Erp 2022 Netherlands [25]	Support needs of Dutch young adult childhood cancer survivors (YACCS)	Quantitative Cross-sectional Survey Self-developed	Medical topics, psychosocial needs, lifestyle behaviors	Survivors: n = 151 Male: n = 58 Female: n = 93	Hematologic cancer: n = 101 CNS tumor: n = 13 Solid tumor: n = 37	Range = 0.4–17 M = 10.5 (SD 4.5)	Range = 6–27 M = 13.6 (SD 3.8)	60 %
Vetsch 2017 Australia, New Zealand [49]	"Forewarned and forearmed": long-term childhood cancer survivors' and parents' information needs and implications for survivorship models of care	Mixed methods Cross-sectional Survey/ Interviews Self-developed	Medical topics, psychosocial needs, lifestyle behaviors	Survivors: n = 322 Male: n = 145 Female: n = 176	Leukemia: n = 131 Lymphoma: n = 49 Brain cancer: n = 33 Other: n = 107	< 16	M = 19.7 (SD 8.8) Range = 5–59	100 %
Vetsch 2020 Australia, New Zealand [50]	Genetics-related service and information needs of childhood cancer survivors and parents: a mixed-methods study	Mixed methods Cross-sectional Survey/ Interviews Self-developed based on the SCNS-SF34	Genetics-related and services	Survivors: n = 404 Male: n = 174 Female: n = 228	Leukemia: n = 169 Lymphoma: n = 58 Solid tumor: n = 158 Other: n = 4	< 16	Range = 5–59 M = 19.1 (SD 8.1)	80 %

[1] For quantitative and quantitative part of mixed-methods studies only.
[2] Studies may focus on information needs in a specific area or report on information needs in general without restricting to one area.
3 Range, mean, or median if reported.
4 The time since diagnosis was calculated by subtracting age at diagnosis from the age of study.

5 The article reported the numbers inconsistently throughout.
6 The full quality assessment can be found in Appendix B.
Abbreviations: CNS: Central Nervous System; M: Mean, Mdn: Median; SD: Standard Deviation; PNET: Primitive Neuro-ectodermal tumor; TYA: Teenage and young adult, YACCS: Young Adult Childhood Cancer Survivors; CCSS-NAQ: Childhood Cancer Survivor Study (CCSS) - Needs Assessment Questionnaire (NAQ); EORTC QLQ-INFO26: European Organization for Research and Treatment of Cancer (EORTC); Quality of Life Questionnaire (QLQ) to assess information provided to cancer patients (INFO26); SCNS-SF34: Supportive Care Needs Survey (SCNS); Short Form (SF34); SUNS: Survivor Unmet Needs Survey (SUNS).

findings were interpreted within the context of their methodological rigor.

2.7. Data synthesis

A narrative synthesis approach was used to systematically categorize and interpret data on information needs [33]. The Childhood Cancer Survivor Study Needs Assessment Questionnaire (CCSS-NAQ) framework [34] initially guided domain grouping, but domain names were refined for accuracy. For example, 'Coping' was renamed 'Psychosocial well-being and support' to reflect the broader psychosocial themes identified in the review, such as psychosocial resources and social-emotional consequences. 'Employment' was broadened to include education for comprehensiveness. The finalized domains and corresponding needs are presented in Table 2.

The software MAXQDA was used to assign needs to domains. This process was conducted by the first author (EB) and checked by the shared first author (AM), with discrepancies resolved through team discussions (EB, AM, KO, AI, GM). Study characteristics were summarized using descriptive statistics, with prevalence rates reported as percentages. When not directly provided, rates were calculated if relevant data were available. Appendix C presents prevalence rates per study, while Table 2 summarizes them as ranges to highlight variations.

Factors associated with information needs were categorized into five groups: socio-demographic, mental health, physical health, cancer diagnosis and treatment, and healthcare and support characteristics. Significant associations (95 % confidence level) are summarized in Table 3, using regression coefficients, correlation values, odds ratios, and p-values. Factors identified through both univariable and multivariable regressions were included.

3. Results

3.1. Results of the search

We identified 1299 articles, of which 200 were duplicates. Title and abstract screening of 1099 records led to the identification and removal of an additional 48 duplicates, while 908 records were excluded for not meeting eligibility criteria. This left 143 records for full-text screening, of which 140 full texts were retrieved and assessed. Ultimately, 21 studies were included in the review [25,35–54]. The identification and selection steps are summarized in the PRISMA flow diagram (Fig. 1).

Table 2
Prevalence of information needs.

Domain	Type of information need	Unmet needs			No and met needs		
		Prevalence range (%)	Sum of study sizes (N)	References	Prevalence range (%)	Sum of study sizes (N)	References
Non-specified information need		51–77	855	[41,49,51]	13–25	1671	[36,38,40,49]
Cancer-related health information	Late effects	19–86	4608	[35-38,40,41,45,46,48,49,54,25]	17–93	655	[35,40,45,46]
	Fertility/ having children after cancer	2–75	1853	[41,42,45,46,49,50,54,25]	58	271	[46]
	Personal and familial genetic predisposition to cancer	38–69	675	[46,50]	61	271	[46]
	Illness, prognosis and relapse	3–68	3058	[36,37,40,41,46,47,49,51]	31–60	590	[40,46]
	Cancer treatment	10–65	3095	[35-37,40,41,46,47,49]	7–82	716	[36,40,46,54]
Follow-up care and prevention	Follow-up consultations to discuss current health	92	112	[54]			
	Long-term follow-up care	2–71	4301	[36-38,40,41,45-47,49,51]	30–58	590	[40,46]
	Preventive self-check actions	10–24	51	[45]			
Healthcare system interactions	Health literacy	31–36	1371	[38,47,48]			
	Transition to adult care	9–24	600	[41,47,51]			
Living a healthy lifestyle	Health promoting behaviors	4–60	1425	[41,45,46,49,54,25]	68–71	271	[46]
	Managing cancer repercussions	19–47	1064	[43,46,47,49]	53–80	675	[43,46]
Psychosocial well-being and support	Psychological support and resources	12–40	891	[41,45,49]			
	Social-emotional consequences of cancer	22–34	422	[46,25]	79	271	[46]
Sexual health	Sexual relationships	< 1–32	1170	[41,47,49,25]			
	Advice on sexual problems	17–24	705	[46,49,54]	83	271	[46]
Finances	Advice on finances	2–50	1106	[41,48,49,25]			
Relationships	Communicating with loved ones	11–20	593	[46,49]	89	271	[46]
	Impact of cancer on social life	2–20	840	[41,49]			
Education and employment	Advice on education and employment	< 1–18	991	[41,49,25]	36	112	[54]
Insurances	Insurances after cancer	28–47	534	[46,54,25]	68	271	[46]
Peer support	Perspectives from other survivors	7–55	1723	[37,49,51]			
	Obtaining future perspectives	35	151	[25]			

Note: Detailed information for each study is available in Appendix B.

Table 3
Factors associated with information needs.

Area	Associated factors	Unmet needs				Met needs			
		Statistical analysis	Coefficient/Result	p-value	Ref.	Statistical analysis	Coefficient/ Result	p-value	Ref.
Socio-demographic characteristics	Older age at study (continuous)	ANOVA	F(2385) = 7.76	$p < 0.001$	[43]				
	Female sex (vs male)	Multivariable logistic regression	OR = 1.87	$p = 0.002$	[41]				
	Education: low level (reference category not reported)	Chi-squared test	$X^2 = 7.32$[1]	$p = 0.026$	[41]				
	Education: post-school qualification (vs no post-school qualification)	Multivariable logistic regression	Genetic-related[2]: OR = 0.48	$p = 0.015$	[50]				
	Ethnicity: not Australian/ New Zealand (vs Australian/ New Zealand ethnicity)	Chi-squared test	Fatigue[2]: $X^2 = 11.03$ Pain[2]: $X^2 = 9.08$	$p = 0.004$ $p = 0.011$	[43]				
Mental health characteristics	Psychological distress (BSI ≥ 57 vs BSI < 57)	Univariable logistic regression	OR = 2.50	$p = 0.025$	[36]				
	Lower mental HRQoL (continuous)	Univariable logistic regression	Disease[2]: OR= 0.96 Treatment[2]: OR = 0.97	$p = 0.006$ $p = 0.041$	[36]				
	Lower mental HRQoL (continuous)	*t*-test	Illness (information need yes vs no)[23]: $M_{diff} = -3.40$ Treatment (information need yes vs no)[23]: $M_{diff} = -3.16$ Late effects (information need yes vs no)[23]: $M_{diff} = -4.94$	$p = 0.020$ $p = 0.030$ $p = 0.006$	[40]				
	Higher fear of cancer recurrence (continuous)	Multiple regression	Pain (unmet information need yes vs no)[23]: B = 0.48 Fatigue (unmet information need yes vs no)[23]: B = 0.29	$p = 0.001$ $p = 0.015$	[43]				
	Higher anxiety (continuous)	*t*-test	Illness (information need yes vs no)[23]: $M_{diff} = 2.95$ Late effects[23]: $M_{diff} = 2.31$	$p = 0.004$ $p = 0.019$	[40]				
	Higher depression (continuous)	*t*-test	Illness (information need yes vs no)[23]: $M_{diff} = 2.17$ Late effects (information need yes vs no)[23]: $M_{diff} = 3.53$	$p = 0.032$ $p = 0.005$	[40]				
Physical health characteristics	Health status: severe disability (vs perfect health)	Multivariable logistic regression	OR = 2.47	$p = 0.006$	[41]	Multivariable logistic regression	OR = 0.30[4]	$p < 0.001$	[41]
	Physical experiences (those reporting physical experiences e.g., endocrine issues, infertility, and pain, vs those who did not)	Chi-squared test	Late effects[2]: $X^2 = 6.318$ Dealing with late effects[2]: $X^2 = 11.323$	$p < 0.008$ $p < 0.001$	[46]				
	Experience of more late effects (reference category not reported)	Chi-squared test	Dealing with symptoms[2]: $X^2 = 11.527$ Late effects[2]: $X^2 = 6.383$ Managing late effects[2]: $X^2 = 9.669$ Dealing with anxiety about risk of recurrence[2]: $X^2 = 7.620$	$p < 0.021$ $p < 0.012$ $p < 0.002$ $p < 0.006$	[46]				

(*continued on next page*)

Table 3 (*continued*)

Area	Associated factors	Unmet needs				Met needs			
		Statistical analysis	Coefficient/Result	p-value	Ref.	Statistical analysis	Coefficient/ Result	p-value	Ref.
Cancer diagnosis and treatment characteristics	Longer time since diagnosis (reference category not reported)	Chi-squared test ANOVA	$X^2 = 9.05$[1] Fatigue[2]: F(2376) = 7.03 Pain[2]: F(2377) = 4.27	p = 0.029 p = 0.001 p = 0.015	[41] [43]				
	Greater time since treatment completion (continuous)	ANOVA	Fatigue[2]: F(2348) = 4.32 Pain[2]: F(2347) = 7.62	p = 0.014 p = 0.001	[43]				
	Shorter time since diagnosis (<10 and 10–15 vs 20 years)					Multivariable logistic regression	< 10 vs > 20 years: OR = 3.13[4] 10–15 vs > 20 years: OR = 2.14[4]	p < 0.001 p = 0.007	[41]
	Diagnosis: CNS tumors (vs all other diagnoses combined)	Chi-squared test	Fatigue[2]: X^2 = 10.59	p = 0.005	[43]				
Healthcare and support characteristics	Reported written information (yes vs no)	Multivariable logistic regression	OR = 0.34	p < 0.001	[41]	Multivariable logistic regression	OR = 3.32[4]	p < 0.001	[41]
	Satisfaction with long-term follow-up care (satisfied vs unsatisfied)	Multivariable logistic regression	Genetic-related[2]: Satisfied: OR = 0.44	p = 0.028	[50]				
	Genetic testing offered and undertaken (yes vs no)	Multivariable logistic regression	Genetic-related[2]: OR = 1.93	p = 0.043	[50]				

1 This univariate association did not remain significant in the multivariate analysis.
2 The association specifically concerns the indicated domain of information needs.
3 In this instance, information need was the outcome rather than the predictor.
4 The outcome of this association was Satisfaction with the extent of information.
Abbreviations: OR, odds ratio; M_{diff}, mean difference; HRQoL, health-related quality of life; CNS, central nervous system; BSI, Brief Symptom Inventory; ANOVA, Analysis of Variance.

3.2. Study characteristics

This review included a total of 5624 participants across 21 studies (range: 14–1386 per study), with manuscripts published between 2009 and 2024. Except for one study from a middle-income country (Brazil), all were conducted in high-income countries: 10 in Europe, seven in North America, and three in Oceania. Detailed characteristics of each included study are presented in Table 1.

The methodologies varied, with 10 quantitative, eight qualitative, and three mixed-methods studies. Most (n = 19) used a cross-sectional design, while only two adopted a longitudinal approach. Data collection methods for qualitative and mixed-methods studies included individual interviews (n = 7), focus groups (n = 3), and write-in sections in a survey for additional information (n = 1). All quantitative studies and quantitative components of mixed-methods studies used questionnaires (n = 13) to assess information needs. Most questionnaires were either self-developed (n = 8) or adapted from existing tools (n = 2). Three studies employed validated instruments: the CCSS-NAQ, the Supportive Care Needs Survey Short Form (SCNS-SF34), and the Survivor Unmet Needs Survey (SUNS).

Only four studies explicitly defined information needs. While most studies included survivors of various cancer types, three focused solely on survivors of central nervous system (CNS) tumors, brain tumors, or lymphomas. The most common diagnoses were hematologic cancers, lymphomas, and CNS tumors. Participant characteristics, including age at study, age at diagnosis, and time since diagnosis, varied across studies.

3.3. Study quality

The quality assessment results are summarized in Table 1, with the full details provided in Appendix B. All studies met the two MMAT screening criteria. The overall methodological quality of the studies was relatively high, with a mean score of 81 % (range: 40 %-100 %). Ten studies achieved the maximum score (100 %), four scored 80 %, five scored 60 %, and two scored 40 %. The most common issues were identified in quantitative studies, where nine out of 13 studies had limitations related to the representativeness of the sample and response rates below 75 %. These issues accounted for most quality scores falling below 100 %.

3.4. Aim 1: Synthesis of information needs

Our review identified a range of information needs categorized into 11 domains: Cancer-related health information (16 studies on unmet needs, 6 studies on met needs), follow-up care and prevention (11 studies on unmet needs, 2 studies on met needs), healthcare system interactions (5 studies on unmet needs, none on met needs), living a healthy lifestyle (8 studies on unmet needs, 2 studies on met needs), psychosocial well-being and support (5 studies on unmet needs, 1 study on met needs), sexual health (6 studies on unmet needs, 1 study on met needs), finances (4 studies on unmet needs, none on met needs), relationships (3 studies on unmet needs, 1 study on met needs), education and employment (3 studies on unmet needs, 1 study on met needs), insurances (3 studies on unmet needs, 1 study on met needs), and peer support (4 studies on unmet needs, none on met needs).

All domains with their prevalence ranges are reported in Table 2, with the prevalence for each study separately presented in Appendix C. The domains are presented below, separated by unmet and met needs. When available, qualitative findings are described to provide deeper insights into the quantitative results, such as reasons for knowledge gaps and preferences regarding how and when CAYACS wanted to receive the information.

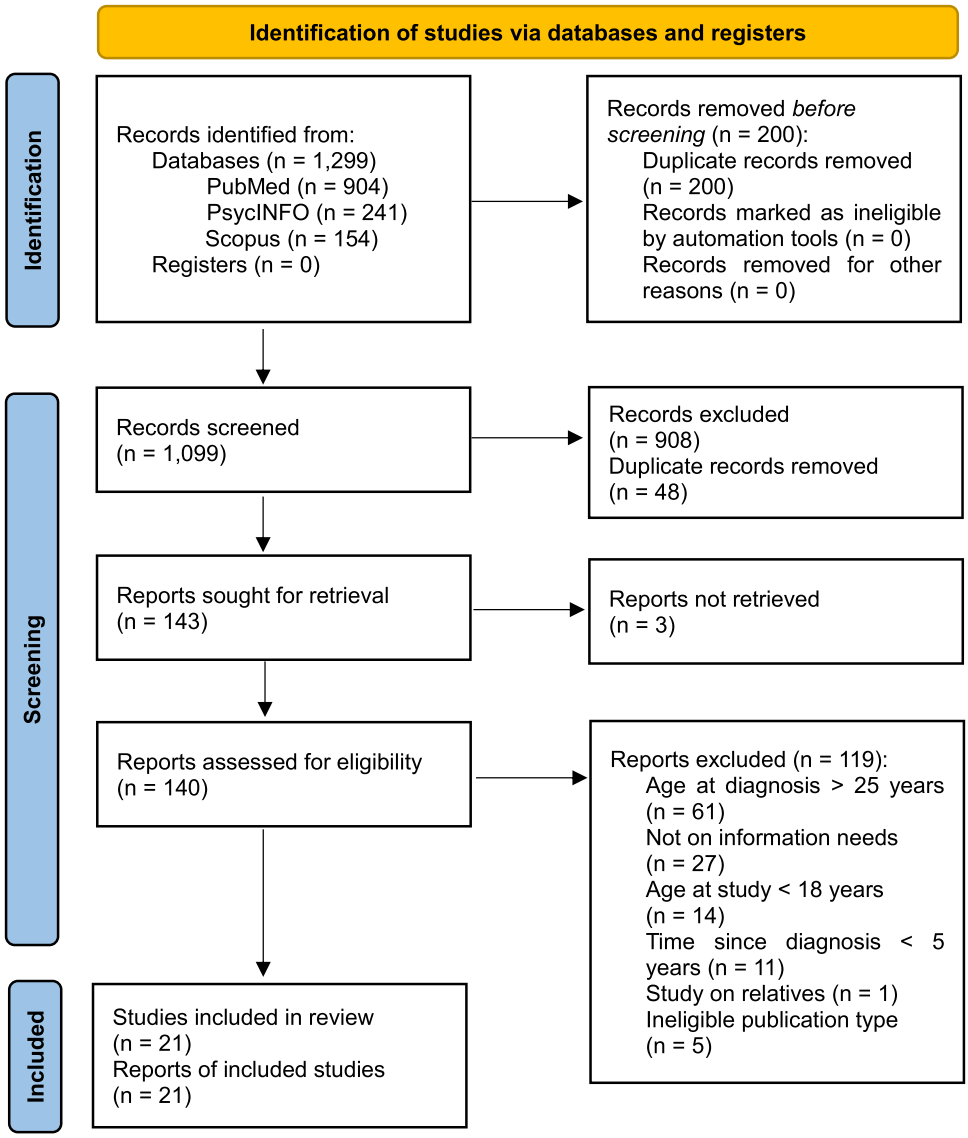

Fig. 1. PRISMA flow diagram of included studies in the systematic review.

3.4.1. Unmet information needs

Unmet information needs were reported in 21 studies across various domains, with prevalences ranging from < 1 %-92 % [25,35–54]. Three studies reported that 51 %-77 % of CAYACS had at least one unmet information need [41,49,51]. In qualitative studies, CAYACS stressed the need for age-appropriate [35,39,42,44,52,53] and personalized [35,44,52,53] information across all stages of the cancer trajectory. Additionally, CAYACS identified reasons for information deficits, such as information being shared with their parents when they were younger, or with themselves as children and therefore they did not remember it [49].

3.4.1.1. Cancer-related health information. CAYACS frequently reported having unmet information needs regarding cancer-related health topics, including late effects (19 %-86 %) [25,35–38,40,41,45,46,48,49,54], fertility or having children after cancer (2 %-75 %) [25,41,42,45,46,49,50,54], personal or familial genetic predispositions to cancer (38 %-69 %) [46,50], illness, prognosis, and relapse (3 %-68 %) [36,37,40,41,46,47,49,51], and cancer treatment (10 %-65 %) [35–37,40,41,46,47,49]. Qualitative studies showed that, although CAYACS reported information about late effects to be overwhelming or distressing, they ultimately appreciated it for enhancing their understanding of their health [35,44,45].

3.4.1.2. Follow-up care and prevention. The prevalence of unmet information needs about follow-up and prevention varied, with 92 % of CAYACS indicating a need for consultations to discuss their current health [51], 2 %-71 % needing information about long-term follow-up care [36–38,40,41,45–47,49,51], and 10 %-24 % about preventive self-checks [45]. CAYACS in qualitative studies stressed the importance of proactive communication about follow-up care, including for those lost to follow-up [53]. They also described that receiving sufficient information about follow-up care empowers them to take charge of it themselves [49].

3.4.1.3. Healthcare system interactions. CAYACS reported unmet information needs regarding healthcare system interactions, including health literacy (31 %-36 %) [38,47,48], such as identifying trustworthy sources of information, knowing where to find additional information, and navigating the transition to adult care (9 %-24 %) [41,47,51]. Qualitative studies have revealed specific challenges with transitioning to adult care, such as a lack of information on where to go, which specialists to see, appointment frequencies, and what medical information to share with new healthcare providers [39,52].

3.4.1.4. Living a healthy lifestyle. CAYACS reported unmet information

needs regarding health-promoting behaviors (4 %-60 %), covering topics such as nutrition, physical activity, sun protection, and alcohol use [25,41,45,46,49,54], as well as regarding the management of cancer repercussions (19 %-47 %), including managing late effects and pain [43,46,47,49]. A qualitative study revealed that CAYACS considered lifestyle advice particularly important for them, and that they preferred receiving it from hospital-based professionals [45].

3.4.1.5. Psychosocial well-being and support. Unmet psychosocial information needs consisted of psychosocial support and resources (12 %-40 %) [41,45,49], and social-emotional consequences of cancer (22 %-35 %), including managing fear of cancer recurrence and reducing stress [25,46]. Qualitative findings emphasized the need for development of coping skills tailored to young adults' challenges [39] and highlighted feelings of "being lost" due to limited available information on accessing formal psychosocial support [53].

3.4.1.6. Relationships. CAYACS reported unmet information needs regarding relationships, including communication about cancer with loved ones (11 %-20 %) [46,49], and the impact of cancer on their social life (2 %-20 %) [41,49]. In a qualitative study, CAYACS expressed a desire to learn how to address their cancer history in social situations, such as answering questions about their cancer history, and managing others' reactions [51].

3.4.1.7. Sexual health. Unmet information needs about sexual relationships varied (<1 %-32 %) and included sexuality and contraception [25,41,47,49]. In addition, information needs regarding advice on sexual problems (17 %-24 %) were reported [46,49,54]. One qualitative study highlighted the sensitivity of the topic, with some CAYACS noting that it remains a subject "no one talks about", despite their need for follow-up on sexual concerns [51].

3.4.1.8. Education and employment. Unmet information needs about education and employment were reported by < 1 %-18 % of CAYACS [25,41,49].

3.4.1.9. Finances. CAYACS reported unmet information needs regarding finances (2 %-50 %), including available financial assistance, how to obtain it, mortgages, and financial advice and support [25,41,48,49]. A qualitative study also revealed the need for information about financial rights in relation to their disease [44].

3.4.1.10. Insurances. Unmet information needs about insurances after cancer (28 %-47 %) included challenges in obtaining or retaining health, life, or disability insurance [25,46,54]. CAYACS in qualitative studies additionally sought information on patient rights and social benefits [44].

3.4.1.11. Peer support. CAYACS reported unmet information needs related to learning from the experiences of other survivors (7 %-55 %) [37,49,51] and future perspectives (35 %) [25]. In a qualitative study, they also mentioned the need for information about existing support or associations [51].

3.4.2. No and met information needs

Less evidence was found for no and met information needs, with only nine out of 21 studies in this review reporting on these needs [35,36,38,40,43,45,46,49,54]. Four studies reported that 13 %-25 % of CAYACS had no needs or had their needs met across all domains [36,38,40,49]. No and met needs were reported across eight domains with varying prevalences: cancer-related health information (7 %-93 %) [35,36,40,45,46,54], follow-up care and prevention (30 %-58 %) [40,46], living a healthy lifestyle (53 %-80 %) [43,46], psychosocial well-being and support (79 %) [46], sexual health (83 %) [46], relationships (89 %) [46], education and employment (36 %) [54], and insurances (68 %) [46].

Several studies reported that CAYACS with met needs had already received information on various topics, such as during long-term follow-up care [40,45]. In qualitative studies, some CAYACS were undecided about how much information they wanted to receive. They described receiving information as a double-edged sword: on one hand, they found it frightening or wished to leave the cancer experience behind; on the other hand, they recognized it as essential for preparing for potential late effects [44,49].

3.5. Aim 2: Factors associated with information needs

Six studies explored factors associated with information needs [36,40,41,43,46,50]. The significant findings are summarized below and presented in detail in Table 3.

3.5.1. Socio-demographic characteristics

Female sex [41], older age at study [43], and a lower educational attainment were associated with more unmet information needs [41]. Lower educational attainment was also specifically associated with more genetic-related information needs [50]. In a study conducted in Oceania, participants of ethnicities other than Australian/New Zealand reported more information needs regarding fatigue and pain [43].

3.5.2. Mental health characteristics

Higher psychological distress was associated with more unmet information needs [36]. Lower mental HRQOL [36,40], higher anxiety [40], and higher depression [40] were associated with greater unmet information needs across various domains, including illness, late effects, and treatment. Additionally, higher fear of cancer recurrence was associated with more information needs about pain and fatigue [43].

3.5.3. Physical health characteristics

CAYACS with a severe disability (vs those reporting perfect health) more frequently had unmet information needs and less frequently had met needs [41]. CAYACS reporting physical issues (e.g., endocrine issues, infertility, and pain, vs those who did not) were more likely to report unmet information needs about late effects and dealing with late effects [46]. Additionally, the experience of more physical late effects was associated with more information needs about dealing with symptoms, late effects, managing late effects, and dealing with anxiety about the risk of recurrence [46].

3.5.4. Cancer diagnosis and treatment characteristics

Longer time since diagnosis [41,43] and treatment completion [43] were associated with more information needs, particularly regarding fatigue and pain. Another study found that CAYACS diagnosed less than 10 years ago, as well as those diagnosed 10–15 years ago, were more satisfied with the information received compared to survivors diagnosed over 20 years ago [41]. Survivors of CNS tumors reported more unmet information needs about fatigue compared to survivors of other cancer types [43].

3.5.5. Healthcare and support characteristics

Not receiving written information was strongly associated with more unmet information needs, while receiving written information was associated with more met needs [41]. Greater satisfaction with long-term follow-up care was associated with fewer genetic-related information needs, whereas having genetic testing offered and undertaken was associated with more genetic-related information needs [50].

4. Discussion and conclusion

This systematic review provided the first comprehensive overview of information needs among adult CAYACS, and the factors associated with

these needs. The review included both quantitative and qualitative findings of 21 studies, involving a total of 5624 CAYACS.

4.1. Discussion

4.1.1. Prevalence of information needs

The findings show that 51 %-77 % of CAYACS report having at least one information need [41,49,51]. These needs spanned 11 domains, highlighting the wide range of information required by CAYACS [25, 35–54]. These finding are consistent with findings in adult cancer survivors, where studies report that 74–84 % experience unmet practical, physical, or emotional needs 1–3 years after treatment [55].

Cancer-related health information consistently emerged as the most dominant need across studies, particularly regarding late effects and fertility; follow-up care and prevention, especially the desire to discuss their current health during follow-up consultations; and living a healthy lifestyle, specifically regarding health-promoting behaviors. Qualitative studies provided insight into reasons for unmet needs, such as information being shared only with parents, or provided at a young age and not being remembered in adulthood [49].

Prevalence rates of needs varied across studies, but direct comparisons were challenging due to methodological differences, measurement instruments, and participant characteristics. For example, needs may differ based on survivors' current life stage [43]. The included studies involved CAYACS of varying ages at study - from young adulthood to late adulthood - and this heterogeneity likely influenced reported information needs. Furthermore, some studies focused on sub-groups of CAYACS, such as CNS tumor survivors [41], or those attending long-term follow-up clinics [45,54].

Notably, studies involving CAYACS attending long-term follow-up clinics reported lowest prevalences of unmet needs in the review (e.g., 2 % unmet needs about infertility and follow-up care) [45]. This may suggest that access to structured survivorship care could be an important explanatory mechanism. Regular, guided contact with clinicians may support survivors' health literacy, helping them ask questions and critically evaluate information. This underscores the importance of LTFU care, though its availability and accessibility remain inconsistent across countries, potentially contributing to variability in information needs [56].

Interestingly, relatively low prevalence rates were reported for information needs on relationships and sexual health, despite the significant challenges CAYACS may face with sexual function after cancer treatment [57]. CAYACS might have been reluctant to express their need for information in these areas. From a coping theory perspective, some survivors may engage in avoidant coping, steering away from potentially distressing information about sensitive topics. Such avoidance may reduce short-term discomfort but could contribute to persistent unmet needs when opportunities for safe and supportive discussion are lacking [58]. A sensitive and proactive approach from healthcare providers could help to address these needs [59].

While 21 studies addressed unmet information needs, only nine studies reported on no and met needs [35,36,38,40,43,45,46,49,54]. The findings show that 13 %-25 % of CAYACS either had no needs or had their needs met across all domains [36,38,40,49]. No and met needs were reported across eight domains with highest prevalences for cancer-related health information, living a healthy lifestyle, sexual health, and relationships. Reasons for reported met or no needs could be that CAYACS had already received the necessary information [40,45]. Additionally, in qualitative studies, some CAYACS reported to be unsure about whether they wanted to receive information, as it could be frightening, and they preferred to leave the cancer experience behind [44,49]. Some may adopt an avoidant coping style, choosing to avoid health information to protect themselves from fear or distress [60]. This highlights the importance of stepwise, personalized delivery of information on late effects that considers the emotional responses CAYACS may experience upon receiving such information. Finally, guidelines for follow-up care lack detailed recommendations on how and when to provide information about late effects. CAYACS may have received information but not fully understood it, as an observational study showed significant differences in how healthcare professionals deliver it [61]. Standardized procedures for educating CAYACS on their medical history and health risks would help address this issue.

The high prevalence of met information needs for cancer-related health information and living a healthy lifestyle contrast with the high prevalence of unmet needs in these domains. Outliers with high prevalence rates of met needs were again observed in the study of survivors attending a long-term follow-up clinic [45], possibly reflecting the recent receipt of information by participants. Another study [43] found a high prevalence of met needs in the lifestyle domain, specifically regarding managing pain, a subdomain where unmet needs were relatively low. Overall, however, met needs were lower than unmet needs.

4.1.2. Associated factors of information needs

CAYACS' characteristics play an important role in their information needs [35,36,38,40,43,45,46,49,54].

Female CAYACS reported more unmet information needs than males, potentially due to greater health awareness or higher levels of health consciousness [62]. Older CAYACS also reported more information needs, which may reflect a heightened sense of responsibility for managing their health as they age [17]. Additionally, the association between lower educational attainment and greater information needs may stem from lower health literacy in this group [14,21], from the greater challenges CAYACS with lower education may face regarding HRQOL [63], or from potential challenges in accessing care.

CAYACS with mental and physical health issues reported more information needs. Symptoms such as pain, fatigue, and depression may amplify worries about late effects and overall health, leading to an increased need for information [17,21,24]. Additionally, mental health issues, such as anxiety and stress, may be exacerbated by the presence of unmet information needs.

Regarding cancer-related medical characteristics, this review shows that longer time since diagnosis and treatment completion are associated with higher information needs. This suggests that needs change over time, as late effects can occur several years after the end of treatment. Therefore, information provision and follow-up care should be adapted over time to meet CAYACS' needs [5,64,65]. Survivors of CNS tumors reported more unmet needs than survivors of other cancers. This may reflect the higher risk this group faces for health issues, lower quality of life, and poor health satisfaction [7,66]. Therefore, particular attention must be devoted to this vulnerable subgroup.

Regarding healthcare and support characteristics, the provision of written information decreased CAYACS' likelihood of having unmet information needs. Written information allows survivors to have the information they need readily available. However, offering and conducting genetic testing may increase genetic-related information needs, possibly due to questions raised during the process or pre-existing interest among those undergoing testing.

4.1.3. Study limitations

Although the overall quality of the studies included in this systematic review was relatively high (average: 81 %), several limitations should be considered when interpreting the results. First, most studies employed a cross-sectional design. Consequently, little is known about how information needs develop over time. However, the identified association between a longer time since diagnosis/treatment and higher information needs suggests that these needs may increase over time. Second, most quantitative studies had limitations regarding the representativeness of their samples, with response rates often below 75 %. As a result, the prevalence rates reported in these studies may differ from those in the broader population of CAYACS. Third, as most studies were conducted in high-income countries, the generalizability of the findings to low- and middle-income countries is limited. Additional research in

these regions is important to develop a better understanding of survivors' information needs in these contexts. Fourth, studies relied on self-reported measures, which reflect survivors' perceptions and may be influenced by social desirability, particularly for sensitive topics such as sexuality. Moreover, many samples were clinic-based, likely overestimating met needs and underestimating unmet needs because these survivors have greater access to structured information. Finally, the included studies were conducted across various countries and among diverse survivor populations, using different methodologies. CAYACS varied in socio-demographic and clinical characteristics, including their age at diagnosis (childhood, adolescence, or young adulthood). While the inclusion of qualitative and quantitative studies with diverse survivor groups adds depth to this review, the prevalence ranges should be interpreted with caution. To address this, we have also reported individual study characteristics (Table 1) and prevalence rates (Appendix C).

4.2. Practice implications

This review highlights the wide range of information needs of CAYACS across various domains. Healthcare providers and policymakers should ensure that CAYACS receive age-appropriate, personalized information. Priority should be placed on high-impact domains identified in this review: (1) late effects and their management, (2) follow-up care pathways and surveillance recommendations, (3) fertility and reproductive health, and (4) lifestyle behaviors.

The observation that unmet information needs were least prevalent among CAYACS attending LTFU clinics highlights their potential role in addressing information gaps. Integrating structured information delivery into routine survivorship pathways, such as annual survivorship consultations, digital treatment summaries, or risk-stratified follow-up protocols, may help standardize information access. Validated tools, such as the CCSS-NAQ, may be used to identify CAYACS with high information and support needs [38]. Previous studies have shown that CAYACS value information personalized to their medical and psychological situation, with a preference for written material [5,14,67,68]. A patient-centered approach that considers the survivors' gender, age, educational level, and cultural background is necessary to reduce the risk of adverse mental health outcomes associated with information needs [12]. Digital, personalized treatment summaries developed by healthcare professionals have been shown to enhance survivors' health awareness, communication with providers, and adherence to follow-up care [69]. The importance of such a tool is underscored by the finding that one of the most frequently reported information needs in this review concerned cancer treatment. However, to implement such programs effectively, it is crucial to understand the electronic health literacy of CAYACS, which warrants further research [69]. Clear, evaluable outcomes (e.g., reductions in anxiety or uncertainty measured via validated instruments) should be used to assess the impact of information interventions.

Future research should explore the effectiveness of age-appropriate communication methods, focusing on preferred communication modes to address unmet information needs, particularly regarding late effects and their management, follow-up care, and lifestyle. Longitudinal studies are needed to assess how information needs evolve over time and determine the optimal timing for information provision. Additionally, examining the feasibility and impact of eHealth tools tailored to CAYACS could offer insights into the digital delivery of personalized information on cancer-related topics. Understanding CAYACS' ability to engage with these tools will be essential for their successful implementation. Although this review has shown that socio-demographic characteristics are associated with greater information needs, further research should explore whether preferences for information provision also differ across groups to better tailor interventions to the specific needs of different age groups.

4.3. Conclusion

Adult CAYACS report high levels of information needs, particularly regarding cancer-related health information (2 %-86 %), follow-up care and prevention (2 %-91 %), and living a healthy lifestyle (4 %-60 %). Female CAYACS, as well as those who are older, have lower educational attainment, poorer mental and physical health, a longer time since diagnosis or treatment, or a CNS tumor diagnosis, report higher information needs and require special attention. To improve CAYACS' quality of life, communication strategies should address the information needs of CAYACS by providing age-appropriate and personalized information throughout the cancer trajectory and into survivorship. Electronic and mobile health tools are promising ways to deliver such support.

Author contributions

[Author contribution statements will be provided in the final accepted version].

CRediT authorship contribution statement

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to assist with language editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Search strategy

	PubMed	PsycINFO	Scopus
Block 1: Information needs	"needs Assessment"[Mesh] OR "information need*" OR "information preference*" OR "need for information" OR "need of information" OR "health information" OR "knowledge need*" OR "wish for information" OR "wish of information" OR "desire for information" OR "desire of information" OR "request for information" OR "requests of information" OR "informational need*" OR "informative need*"	"needs assessment" OR "information need*" OR "information preference*" OR "need for information" OR "need of information" OR "health information" OR "knowledge need*" OR "wish for information" OR "wish of information" OR "desire for information" OR "desire of information" OR "request for information" OR "requests of information" OR "informational need*" OR "informative need*"	{need* assessment} OR {information need*} OR {information preference*} OR {need for information} OR {need of information} OR {health information} OR {knowledge need*} OR {wish for information} OR {wish of information} OR {desire for information} OR {desire of information} OR {request* for information} OR {informational need*} OR {informative need*}
Block 2: Survivors	"cancer survivors"[Mesh] OR "cancer survivor" OR "survivors, cancer" OR "survivors of childhood cancer" OR "childhood cancer survivor*" OR "cancer survivorship" OR "long-term cancer survivor*" OR "long term cancer survivor*" OR "Survivors"[Mesh] OR "long-term survivor*" OR "long term survivor*" OR "survivor, long-term" OR "survivors, long-term" OR surviv*	"cancer survivor*" OR "survivors, cancer" OR "survivors of childhood cancer" OR "childhood cancer survivor*" OR "cancer survivorship" OR "long-term cancer survivor*" OR "long term cancer survivor*" OR surviv* OR "long-term survivor*" OR "long term survivor*" OR "survivor, long-term" OR "survivors, long-term"	{cancer survivor*} OR {survivor*, cancer} OR {survivor* of childhood cancer} OR {childhood cancer survivor*} OR {long-term cancer survivor*} OR {long term cancer survivor*} OR surviv* OR {long-term survivor*} OR {long term survivor*} OR {survivor*, long-term}
Block 3: Cancer	"neoplasms"[Mesh] OR tumor* OR tumour* OR neoplas* OR cancer* OR "malignant neoplasm*" OR malignanc* OR oncolog* OR carcinoma* OR leukemia* OR leukaemia* OR lymphoma* OR medulloblastoma* OR sarcom* OR "Radiotherapy"[Mesh] OR radiotherap* OR "Chemoradiotherapy"[Mesh] OR "Chemotherapy, Adjuvant"[Mesh] OR chemotherap*	neoplas* OR tumor* OR tumour* OR cancer* OR "malignant neoplasm*" OR malignanc* OR oncolog* OR carcinoma* OR leukemia* OR leukaemia* OR lymphoma* OR medulloblastoma* OR sarcom* OR radiotherap* OR chemoradiotherap* OR "chemotherapy adjuvant" OR chemotherap*	neoplasm* OR tumor* OR neoplas* OR cancer* OR {malignant neoplasm*} OR malignanc* OR oncolog* OR carcinoma* OR leukemia* OR leukaemia* OR lymphoma* OR medulloblastoma* OR sarcom* OR radiotherap* OR chemoradiotherap* OR {adjuvant chemotherapy} OR chemotherap*
Block 4: Childhood, adolescence, young adulthood	"child"[Mesh] OR "Adolescent"[Mesh] OR child* OR adolescen* OR teen* OR teenager* OR youth* OR kid OR kids OR "Child, Preschool"[Mesh] OR "preschool child" OR "Infant"[Mesh] OR "Young Adult"[Mesh] OR "young adult*" OR infant* OR newborn* OR neonat* OR pediatric* OR paediatric* OR girl* OR boy OR boys OR toddler* OR "pre-schooler*" OR preschooler* OR "pre schooler*" OR AYA OR AYAC* OR CAYA* OR TYA	child* OR adolescen* OR teen* OR teenager* OR youth* OR kid OR kids OR "child, preschool" OR "preschool child" OR infant* OR "young adult*" OR infant* OR newborn* OR neonat* OR pediatric* OR paediatric* OR girl* OR boy OR boys OR toddler* OR "pre-schooler*" OR preschooler* OR "pre schooler*" OR AYA OR CAYAC OR CAYACS OR CAYA* OR TYA	child* OR adolescen* OR teen* OR teenager* OR youth* OR kid* OR {preschool child*} OR infant* OR {young adult*} OR newborn* OR neonat* OR pediatric* OR paediatric* OR girl* OR boy* OR toddler* OR {pre-schooler*} OR preschooler* OR {pre schooler *} OR AYA OR AYAC* OR CAYA* OR TYA

Appendix B. Results of the quality appraisal of included publications according to the Mixed Methods Appraisal Tool (MMAT), version 2018

	Qualitative					Quantitative descriptive					Mixed methods					Score
First author, year	1.1	1.2	1.3	1.4	1.5	4.1	4.2	4.3	4.4	4.5	5.1	5.2	5.3	5.4	5.5	
Baudry et al., [70]	Y	Y	Y	Y	Y											100 %
Billman et al., [71]	Y	Y	Y	Y	Y											100 %
Christen et al., [72]						Y	N	Y	N	Y						60 %
Claridy et al., [73]						Y	Y	Y	N	Y						80 %
Cox et al., [74]						Y	Y	Y	N	Y						80 %
D'Agostino, Edelstein [75]	Y	Y	Y	Y	Y											100 %
Frederick et al., [76]	Y	Y	Y	Y	Y											100 %
Gianinazzi et al., [77]						Y	N	Y	N	Y						60 %
Hendriks et al., [78]	Y	Y	Y	Y	Y											100 %
Hoven et al., [79]						Y	Y	Y	Y	Y						100 %
Jardim et al., [80]	Y	Y	Y	Y	Y											100 %
Kelada et al., [81]						Y	C	Y	N	Y						60 %
Lie et al., [82]	Y	Y	Y	Y	Y											100 %
Mayes et al., [83]	Y	C	C	C	Y											40 %
McClellan et al., [84]	*Y*	*N*	*Y*	*Y*	*C*	*Y*	*N*	*Y*	*N*	*Y*	Y	Y	Y	Y	N	80 %
Michel et al., [85]						Y	Y	Y	Y	Y						100 %
Nicklin et al., [86]						C	N	Y	N	Y						40 %
Park et al., [87]						Y	N	Y	N	Y						60 %
van Erp et al., [88]						Y	N	Y	N	Y						60 %
Vetsch et al., [89]	*Y*	*Y*	*Y*	*Y*	*Y*	*Y*	*C*	*Y*	*C*	*Y*	Y	Y	Y	Y	Y	100 %
Vetsch et al., [90]	*Y*	*Y*	*Y*	*Y*	*Y*	*Y*	*N*	*Y*	*N*	*Y*	C	Y	Y	Y	Y	80 %

Notes:
Abbreviations: Y, yes; N, no; C, can't tell.

Screening questions (for all types):
S1. Are there clear research questions?
S2. Do the collected data allow to address the research questions?
Please note: Further appraisal may not be feasible or appropriate when the answer is 'No' or 'Can't tell' to one or both screening questions
Qualitative studies:
1.1 Is the qualitative approach appropriate to answer the research question?
1.2 Are the qualitative data collection methods adequate to address the research question?
1.3 Are the findings adequately derived from the data?
1.4 Is the interpretation of results sufficiently substantiated by data?
1.5 Is there coherence between qualitative data sources, collection, analysis and interpretation?
Quantitative descriptive studies:
4.1 Is the sampling strategy relevant to address the research question?
4.2 Is the sample representative of the target population?
4.3 Are the measurements appropriate?
4.4 Is the risk of nonresponse bias low?
4.5 Is the statistical analysis appropriate to answer the research question?
Mixed methods studies:
5.1 Is there an adequate rationale for using a mixed methods design to address the research question?
5.2 Are the different components of the study effectively integrated to answer the research question?
5.3 Are the outputs of the integration of qualitative and quantitative components adequately interpreted?
5.4 Are divergences and inconsistencies between quantitative and qualitative results adequately addressed?
5.5 Do the different components of the study adhere to the quality criteria of each tradition of the methods involved?

Appendix C. Identified information needs, according to domain and type

Domain of need	Type of need	Unmet needs			No and met needs		
		Prevalence	Study N	Ref.	Prevalence	Study N	Ref.
Not specified information needs		76.7 %	322	[89]	25 %	1189	[74]
		60 %[2]	15	[70]	23.3 %	322	[89]
		51 %	518	[79]	16.3 %	160	[72]
				[77]	13 %	319	[77]
Cancer-related health information	Potential late effects that might resultfrom treatment	85.7 %	112	[85]	93 %[2]	51	[83]
		80 %	319	[77]	48 %	271/272	[84]
		78.7 %	160	[72]	28.6 %[1,2]	14	[71]
		57.5 %	322	[89]		319	[77]
		57.1 %[1,2]	14	[71]	17 %		
		54 %[3]	151	[88]			
		52 %	271/272	[84]			
		41.5 %	518	[79]			
		19.4 %[2]	51	[83]			
		-[2]	22	[75]			
		-[2]	34	[82]			
	Fertility	75 %-[2]	24	[80]			
		60 %[3]	151	[88]			
		56 %[3]	112	[85]			
		45 %[3]	322	[89]			
		6 %	518	[79]			
		2 %[1,2]	51	[83]			
		-2	22	[75]			
	Things they can do to avoid future health problems	73 %	1386	[73]			
	Genetics-related	69.2 % (With possible genetic conditions) 40.9 % (Without genetic conditions)	404	[90]			
	Illness	68 %	319	[77]	31 %	319	[77]
		58.1 %	160	[72]			
		20.9 %	518	[79]			
		-[2]	22	[75]			
	Cancer treatment (including benefits and side effects before choosing them)	65 %	1386	[73]	32 %	319	[77]
		64 %	319	[77]	7.1 %[1,2]	14	[71]
		55.6 %	160	[72]			
		50 %[1,2]	14	[71]			
		38 %[3]	322	[89]			
		16.4 %	67	[89]			

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Domain of need	Type of need	Unmet needs			No and met needs		
		Prevalence	Study N	Ref.	Prevalence	Study N	Ref.
		-[2]	16	[76]			
		-[2]	22	[75]			
	Kind of cancer or related illness they had	62 %	1386	[73]			
	Cancer genetic risk for other family members	61.5 % (With possible genetic conditions)	404	[90]			
		38.0 % (Without genetic conditions)					
	Being able to ask an expert questions about symptoms that concern them	61 %	1386	[74]			
	Reassurance about their health	57 %	1386	[74]			
	Development of second cancers	53.4 %	322	[89]			
	Cancer-related information needs-[4]	51 %	1189	[74]			
	Signs of cancer and when to be concerned	50.0 %	115	[87]			
	Relapse/ cancer recurrence	47 %[3]	322	[89]			
		13.3 %[2]	15	[70]			
		2.6 %	518	[79]			
	Having children after cancer treatment	46.2 % (With possible genetic conditions)	404	[90]	58 %	271/272	[84]
		42 %	271/272	[84]			
		32.1 % (Without genetic conditions)	404	[90]			
	Decreasing the risk of having cancer again	41 %	271/272	[84]	59 %	271/272	[84]
	Relevant symptoms to report to physicians	40 %	271/272	[84]	60 %	271/272	[84]
		-2	22	[75]			
	Cancer risks to the family	39 %	271/272	[84]	61 %	271/272	[84]
	Melanoma/skin cancer	26 %[3]	322	[89]			
	Complementary and alternative treatments	18 %	271/272	[84]	82 %	271/272	[84]
	Causes of illness	12 %	518	[79]			
	Cancer which is under control or diminishing	11.9 %	67	[86]			
	Treatment/medication-related issues	10.3 %	518	[79]	62 %[3,5]	112	[85]
	Treatment options for late effects	-[2]	34	[82]			
	Prognosis	-[2 3]	22	[75]			
	What survivors need to be aware of, given disease and treatment history, and how to address future concerns	-[2]	16	[76]			
Follow-up care and prevention	Discussion of Current health during follow-up consultation	91.5 %	112	[85]			
	Follow-up care	71.3 %	160	[72]	30 %	319	[77]
		67 %	319	[77]			
		45 %[3]	322	[89]			
		20 %[2]	15	[70]			
		9.4 %	518	[79]			
	Screening tests their doctor might recommend	61 %	1386	[73]	58 %	271/272	[84]
		42 %	271/272	[84]			
	Surveillance-related information needs-[2 4]	33 %	1189	[74]			
	Checking lumps and bumps	24 %[2]	51	[83]			
	Explanations of those tests for which you would like explanations	23.9 %	67	[86]			
	Checking moles and skin changes	10 %[2]	51	[83]			
	Getting regular health check-ups	2 %[1,2]	51	[83]			
	Where to find specialized, individualized long-term follow-up care.	-[2]	28	[78]			
	Prevention	-[2]	22	[75]			
Healthcare system interactions	Knowing which sources of information to trust	36.0 %	115	[87]			
	Health care system information needs[4]	35 %	1189	[74]			
	To be informed about things you can do to help yourself get well	31.3 %	67	[86]			
	To be given written information about the important aspects of your care	23.9 %	67	[86]			
	Test results as soon as feasible	23.9 %	67	[86]			

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Domain of need	Type of need	Unmet needs			No and met needs		
		Prevalence	Study N	Ref.	Prevalence	Study N	Ref.
	Comprehensive treatment summary with survivorship care guidelines for personal benefit and for new providers	13.3 %[2]	15	[70]			
	Rehabilitation	-2	16	[76]			
	Identifying specific information and/or materials that they should communicate to new providers detailing their medical history	9.4 %	518	[79]			
	Skills necessary to effectively access adult health services	-[2]	16	[76]			
	Where to get further information	-[2]	34	[82]			
Living a healthy lifestyle	Lifestyle	60 %[3]	151	[88]			
	Current health behaviors	54 %[3]	112	[85]			
	Late effects management	47 %	271/272	[84]	53 %	271/272	[84]
	Nutrition	35 %[3]	322	[89]	68 %	271/272	[84]
		32 %	271/272	[84]			
		2 %[1,2]	51	[83]			
	Managing fatigue	35 %[3]	322	[89]	67 %[3,6]	404	[81]
		32.2	404	[81]			
	Managing vitamin D	34 %[3]	322	[89]			
	Physical activity	32 %[3]	322	[89]	71 %	271/272	[84]
		29 %	271/271	[84]			
		8 %[2]	51	[83]			
	Managing illness and side-effects at home	20.9 %	67	[86]			
	Pain	20 %[3]	322	[89]	80 %[3,6]	404	[81]
		18.8 %	404	[81]			
	Sun protection	18 %[3]	322	[89]			
		8 %	51	[83]			
	Future health behavior	16 %[3]	322	[89]			
	Information in later life	8.5 %	518	[79]			
	Self-care/care-management	6 %	518	[79]			
	Smoking cessation	4 %[2]	51	[83]			
	Alcohol	4 %[12]	51	[83]			
	What is "normal'' for survivors in terms of physical and psychological functioning	-[2]	34	[82]			
Psychosocial well-being and support	Psychological support	40 %[3]	322	[89]			
		12.4 %	518	[79]			
	Social-emotional consequences of cancer	34 %[3]	151	[88]			
	Managing anxiety about recurrence	22 %	271/272	[84]	79 %	271/272	[84][7]
	Reducing stress	16 %[2]	51	[83]			
	Information on finding formal psychological support	-[2]	28	[78]			
	Psychosocial resources	-[2]	22	[75]			
Sexual health	Contraception	32 %[3]	112	[89]			
		18 %3	322				
	(Dealing with) sexual problems	24 %[3]	112	[85]	83 %	271/272	[84]
		23 %[3]	322	[89]			
		17 %	271/272	[84]			
	Sexual relationships	22.4 %	67	[86]			
	Sexuality	6 %	518	[79]			
		< 1 %	151	[88]			
Finances	Finding what type of financial assistance is available and how to obtain it	50.0 % (Middle childhood diagnosis) 32.0 % (Early childhood diagnosis)	115	[87]			
	Mortgages	47 %[3]	151	[88]			
	Financial advice	17 %[3]	322	[89]			
	Financial support	1.7 %	518	[79]			
	Financial rights	-[2]	34	[82]			
Relationships	Support family members	20 %[3]	322	[89]			
	Relationship issues	20 %[3]	322	[89]			
	Talking about cancer with loved ones	11 %	271/272	[84]	89 %	271/272	[84]
	Consequences for social life	1.7 %	518	[79]			
Education and employment	Advice on education and employment	18 %[3]	322	[89]	36 %[3,5]	112	[85]
		3 %	518	[79]			
		< 1 %	151	[88]			
Insurances	Getting or retaining health, life, or disability insurance after cancer	47 %[3]	151	[88]	68 %	271/272	[84]

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Domain of need	Type of need	Unmet needs			No and met needs		
		Prevalence	Study N	Ref.	Prevalence	Study N	Ref.
		32 %	271/272	[84]			
		28 %[3]	112	[85]			
	Patient rights/social benefits	-[2]	34	[82]			
Survivors' insights	Hearing stories about people with health histories like their own	55 %	1386	[73]			
	Future perspective	35 %[3]	151	[88]			
	Community support	8 %[3]	322	[89]			
	Information about existing support or associations	6.7 %[2]	15	[70]			

[1] Prevalence rates calculated by reviewer with relevant numbers available in the article
[2] Qualitative information
[3] Prevalence rates were interpreted from a bar graph and may not be completely accurate
[4] Domains mentioned in general
[5] Percentage of survivors who discussed this topic during follow-up was higher than the percentage of survivors who wanted to discuss the topic
[6] Managing fatigue: 8 % met need and 59 % no need, managing pain: 8 % met need and 72 % no need
[7] In this study, conflicting participant numbers were reported in the text and table (271 and 272). In the summary table, we have used n = 271.

Data availability

The protocol is available on Prospero, and the data that support the study findings are available upon justified request from the corresponding author.

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